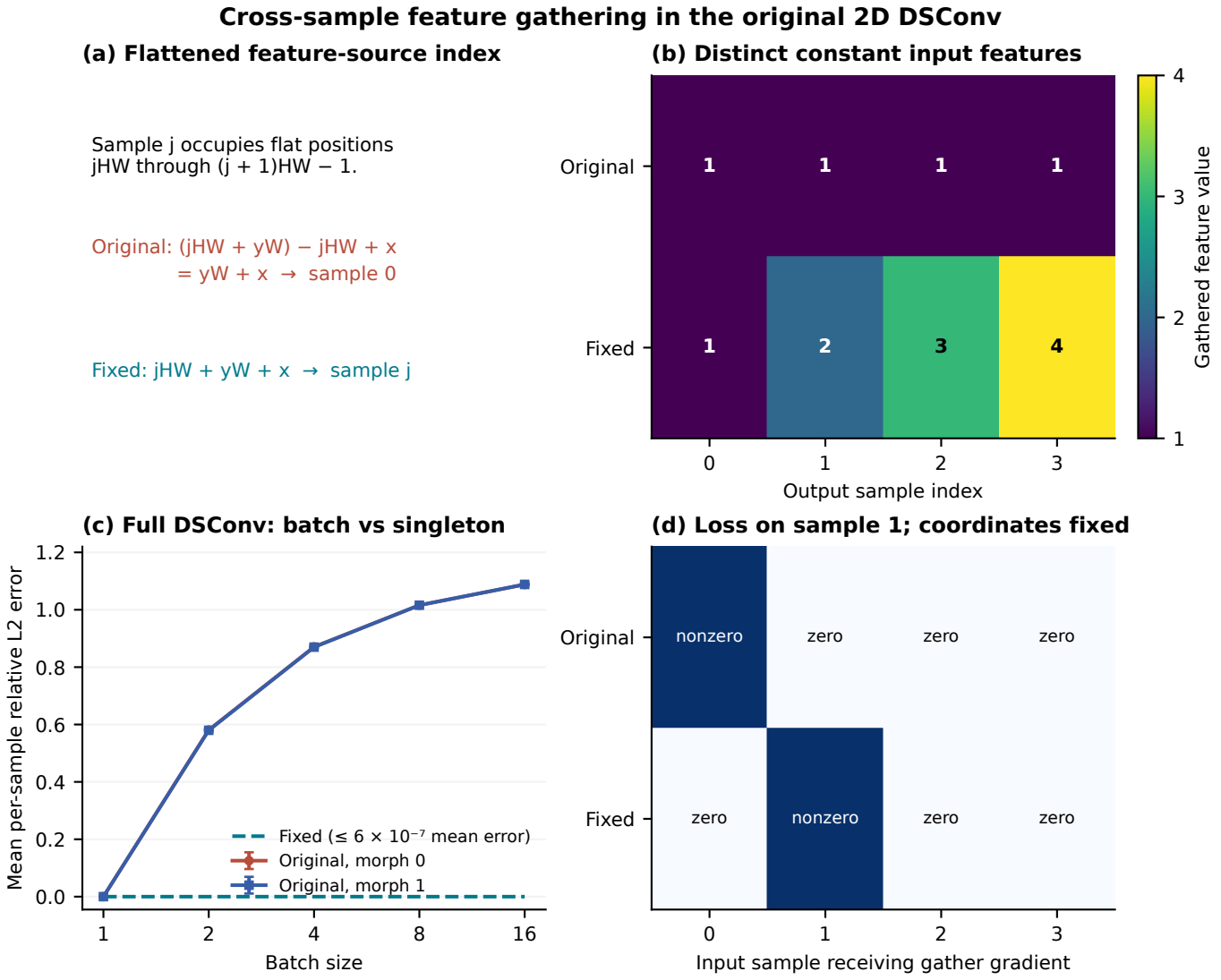
Cross-sample feature gathering in the original 2D DSConv
(a) Flattened feature-source index
Sample j occupies flat positions
jHW through (j + 1)HW − 1.
Original: (jHW + yW) − jHW + x
= yW + x → sample 0
Fixed: jHW + yW + x → sample j
(b) Distinct constant input features
Original
Fixed
1
1
1
1
1
2
3
4
0
1
2
3
Output sample index
Gathered feature value
4
3
2
1
(c) Full DSConv: batch vs singleton
Mean per-sample relative L2 error
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Fixed (≤ 6 × 10⁻⁷ mean error)
Original, morph 0
Original, morph 1
1
2
4
8
16
Batch size
(d) Loss on sample 1; coordinates fixed
Original
Fixed
nonzero
zero
zero
zero
zero
nonzero
zero
zero
0
1
2
3
Input sample receiving gather gradient